

Alcohol Content and Red Wine Quality: A Simple Linear Regression Model

**Shreyes Balaji, Marwan Hegab, Mazin Hussein, Mikael Rotberg, Roberto Rubio,
Jordan Woda**

02/18/2026

Introduction

In this study, we will use a simple linear regression model to analyze and predict red wine quality using alcohol content (% by volume) as the predictor. For our model, we will be using the UC Irvine Machine Learning Repository Wine Quality dataset. The data set provides data on both red and white wine, however for the purposes of this project we will specifically focus on red wine. A simple linear regression models the relationship between one independent variable and one dependent variable. In regards to our data, wine quality is the dependent variable and alcohol content is the independent variable. We will investigate whether alcohol concentration has a significant influence in the wine's quality rating. The dataset contains measurements of alcohol content in red and white wines, as well as sensory quality scores given by experts.

We completed the analysis in R using tidyverse (data import/wrangling and plotting), ggpubr (plot formatting), car (influence diagnostics via Cook's distance), and broom (clean model summaries). Using a 70/30 train–test split (1,119/480), alcohol and quality showed a moderate positive correlation (0.476), and after removing influential training points the final fitted model was $\text{quality}_{\text{hat}} = 1.3817 + 0.4131(\text{alcohol})$. On the test set, the model achieved RMSE = 0.706 and MAE = 0.551, and for a wine with 12.0% alcohol it predicted mean quality 6.339 (95% CI [6.268, 6.410]; 95% PI [5.141, 7.537]).

The goal of this project is to quantify and evaluate the relationship between alcohol content and red wine quality using simple linear regression, then assess performance on a held-out test set and use the fitted model to make predictions of quality.

Data description

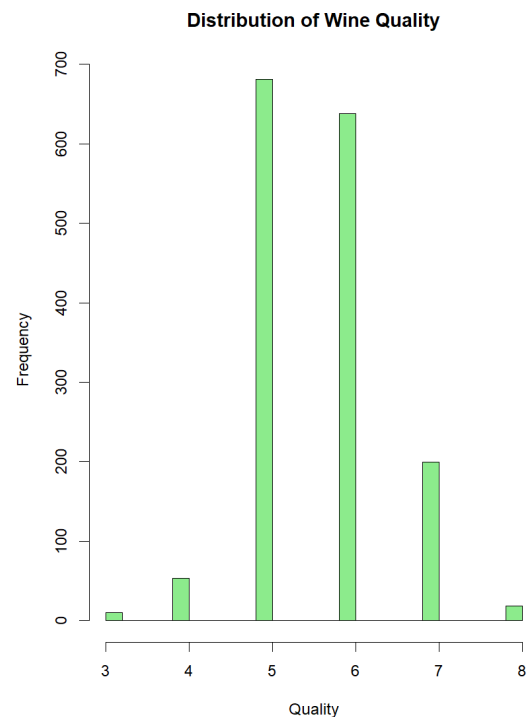
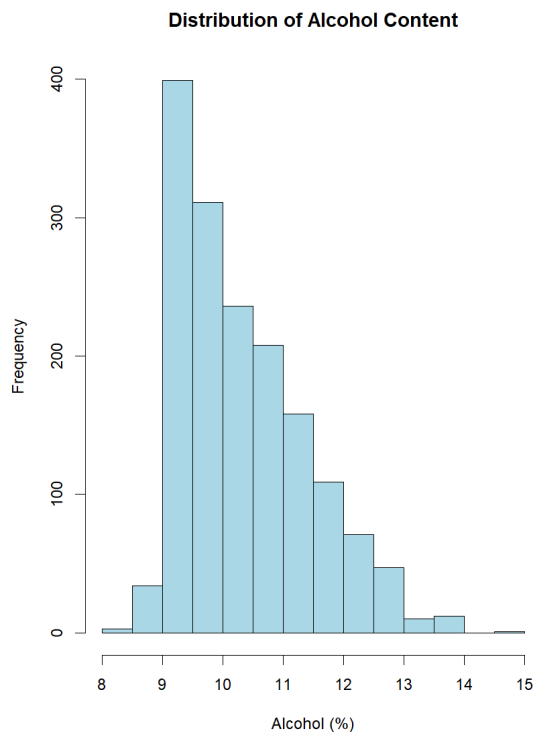
The data set contains 12 columns and 1599 rows. The twelve column names are as follows: "fixed acidity", "volatile acidity", "citric acid", "residual sugar", "chlorides", "free sulfur dioxide", "total sulfur dioxide", "density", "pH", "sulphates", "alcohol", and "quality". All columns are double-precision floating-point numbers, even if some of these don't contain entries with decimal parts. After inspecting the data with the nested functions "colSums(is.na())" and "sum(is.na())", no missing values were found, so no observations were removed for missingness.

Influential points were assessed later on the training set using Cook's distance; 49 influential training observations were removed before fitting the final model.”. Using the IQR method for finding potential outliers in wine quality, the interquartile range (IQR) was calculated to be 1.0. Therefore, based on the $1.5 \times \text{IQR}$ rules, values below 3.5 and above 7.5 were identified as potential outliers. Even though the minimum and maximum observed values were 3.0 and 8.0, respectively, these values are in the valid range of the quality variable and represent legitimate observations. Therefore, those values were retained. The IQR method was similarly applied to the alcohol content variable, and values below 7.1 and above 13.5 were considered potential outliers using the $1.5 \times \text{IQR}$ rule. However, the maximum value of 14.9 falls within a reasonable range for wine alcohol content, so these observations were retained.

Throughout this analysis, quality is the dependent variable and alcohol content is the independent variable. Because this is a simple linear regression, we model quality using alcohol as the single predictor. The table below lists the variables and includes a summary for each.

Variable	Description	min/max/mean	Units
<i>Fixed Acidity</i>	Non-volatile acids (mainly tartaric)	4.6 / 15.9 / 8.32	<i>g/L</i>
<i>Volatile Acidity</i>	Volatile acids (mainly acetic)	0.12 / 1.58 / 0.5278	<i>g/L</i>
<i>Citric Acidity</i>	Citric acid content	0.0 / 1.0 / 0.271	<i>g/L</i>
<i>Residual Sugar</i>	Sugar remaining after fermentation	0.9 / 15.5 / 2.539	<i>g/L</i>
<i>Chlorides</i>	Salt content	0.012 / 0.611 / 0.08747	<i>g/L</i>
<i>Free sulfur dioxide</i>	Free SO ₂ (antimicrobial/antioxidant)	1.0 / 72.0 / 15.87	<i>ppm</i>
<i>Total sulfur dioxide</i>	Total SO ₂ (free + bound)	6.0 / 289.0 / 46.47	<i>ppm</i>
<i>Density</i>	Wine density	0.9901 / 1.0037 /	<i>g/mL</i>

		0.9967	
<i>pH</i>	Acidity/alkalinity measure	2.74 / 4.01 / 3.311	<i>pH scale</i>
<i>Sulphates</i>	Sulphate content (contributes to SO ₂)	0.33 / 2.0 / 0.6581	<i>g/L</i>
Alcohol	Alcohol by volume	8.4 / 14.9 / 10.42	%
<i>Quality</i>	Expert quality score	3.0 / 8.0 / 5.636	<i>score (0–10)</i>



Analysis

Train/test split and data sizes:

The red wine quality dataset contains 1,599 observations and 12 variables. For this project we used a simple linear regression model with wine quality as the response variable (quality) and alcohol content as the predictor variable (alcohol, % by volume). After checking the dataset using `colSums(is.na())`, no missing values were found for alcohol or quality, so no rows were removed due to missing data. The dataset was then split into a training set and a test

set using a reproducible random split in R (set.seed(456)). A 70/30 split was used, which produced 1,119 observations in the training set and 480 observations in the test set.

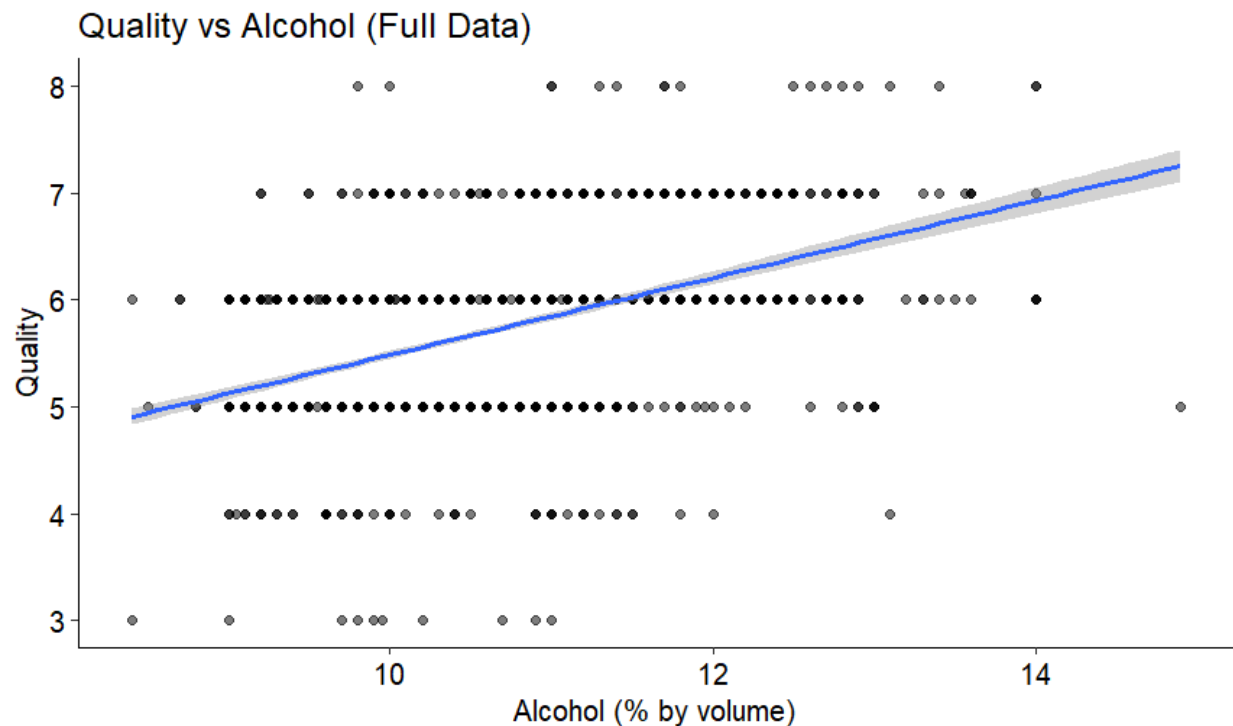


Figure 1. Scatterplot of quality versus alcohol for the full dataset, including the fitted regression line and 95% confidence band.

Initial relationship between alcohol and quality:

The scatterplot in Figure 1 shows a positive relationship between alcohol content and wine quality. As alcohol increases, the average quality score tends to increase. The correlation between alcohol and quality in the full dataset was 0.476, which supports using alcohol as a predictor in a simple linear regression model.

Regression model fit on the training set (initial fit):

Using the training set, the initial fitted regression model was:

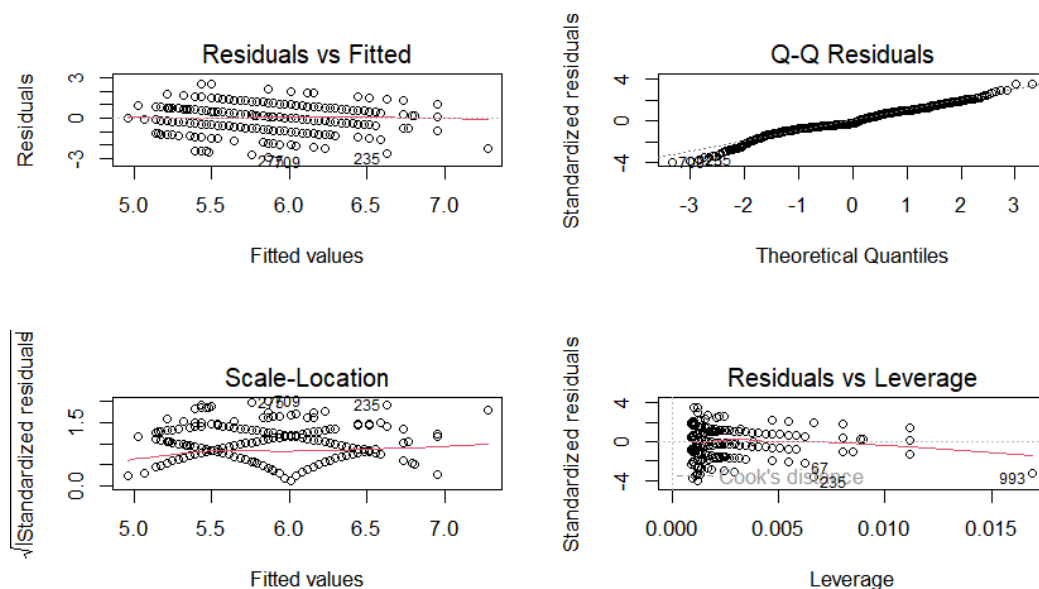
$$\text{quality_hat} = 1.8744 + 0.3627(\text{alcohol})$$

This means that for each 1% increase in alcohol content, the model predicts an average increase of about 0.363 points in the quality score.

Diagonal plot and assumption checks:

To verify the regression assumptions, the standard diagnostic plots were examined (Figure 2). The required diagonal plot check was completed using the Normal Q–Q plot (Figure 3). In the Q–Q plot, the residuals do not follow the diagonal line perfectly, especially in the tails, which suggests the residuals are not perfectly normal. This is not unexpected because the quality response variable takes only integer values (3 to 8), which can create structured residual patterns rather than a smooth normal distribution.

The Residuals vs Fitted plot shows some banding and mild structure, which again is consistent with an integer-valued response. The Scale-Location plot suggests the variance is not perfectly constant across fitted values, but the differences are not extreme. Overall, the diagnostics suggest the linear model is reasonable as a first ap



proximation, but the assumptions are only approximately satisfied.

Figure 2. Standard regression diagnostic plots for the training-set model (Residuals vs Fitted, Normal Q–Q, Scale-Location, and Residuals vs Leverage).

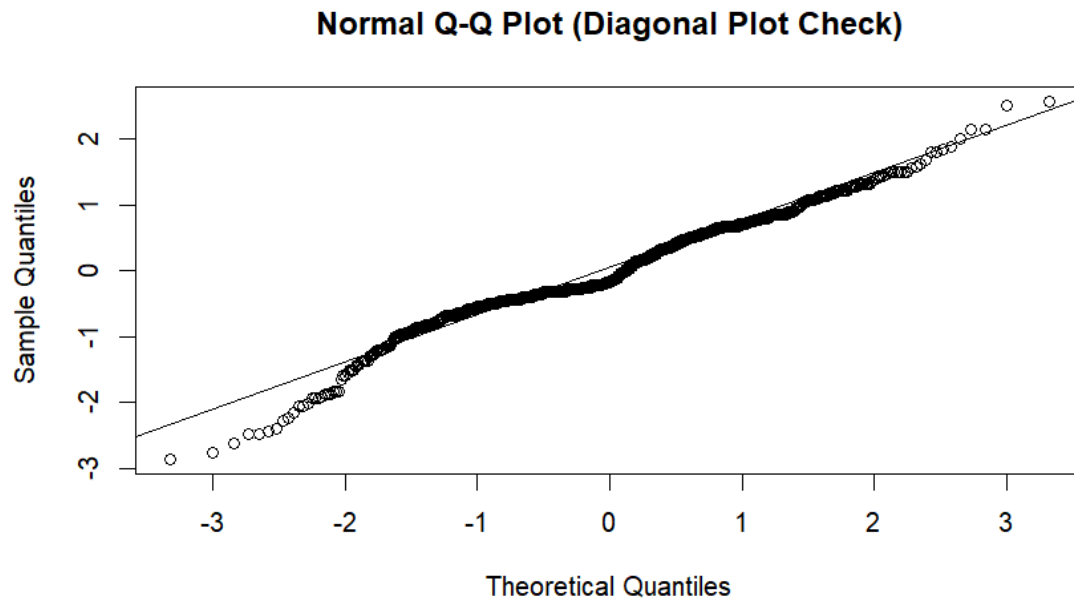


Figure 3. Normal Q–Q plot of residuals (diagonal plot requirement) used to assess whether residuals are approximately normally distributed.

Influential points and model cleaning:

The project instructions require addressing influential points. To do this, Cook's distance was computed for the training-set model. A common cutoff rule is to flag points with Cook's distance $D_i > 4/n$, where n is the training sample size. With $n = 1,119$, the cutoff is:

$$4/1119 = 0.00357$$

Using this rule, 49 training observations were flagged as influential. These 49 points were removed from the training set and the model was refitted on the remaining training observations to obtain the final model used in the evaluation and prediction section. The test set was not changed.

Model evaluation and prediction

Final model after removing influential points:

After removing influential training points, the fitted regression model became:

$$\text{quality_hat} = 1.3817 + 0.4131(\text{alcohol})$$

In this final model, the slope indicates that a 1% increase in alcohol content is associated with an average increase of about 0.413 points in predicted quality. For example, with 12% alcohol the quality is 6.3389.

Model assessment:

The standard regression output from `summary(lm)` provides the main tests needed for model assessment in a simple linear regression.

1. Slope t-test (alcohol coefficient): The alcohol coefficient was statistically significant ($p < 2e-16$). This supports that alcohol is a meaningful predictor of wine quality in this simple model.
2. Overall F-test: The overall regression F-statistic was 485.8 with $p < 2e-16$, showing that the regression model is a statistically significant improvement over an intercept-only model.

For the final model fit on the cleaned training set, the residual standard error (RSE) was 0.609, and R-squared was 0.313 (adjusted R-squared 0.312). This means alcohol alone explains about 31% of the variation in quality in the cleaned training set. The remaining variation is expected because wine quality depends on many chemical variables, not only alcohol.

Model accuracy on the test set:

The final model was evaluated using the test set (480 observations). Predictions were generated for the test set and accuracy was measured using common error metrics:

RMSE = 0.706 (typical size of prediction error on the test set)

MAE = 0.551 (average absolute prediction error)

Test-set R-squared = 0.215

These results show that the model captures a real positive trend, but it does not predict individual wine quality scores with very high precision. The Actual vs Predicted plot (Figure 4) shows noticeable scatter around the 45-degree line, which supports the idea that alcohol alone is not sufficient to fully explain quality.

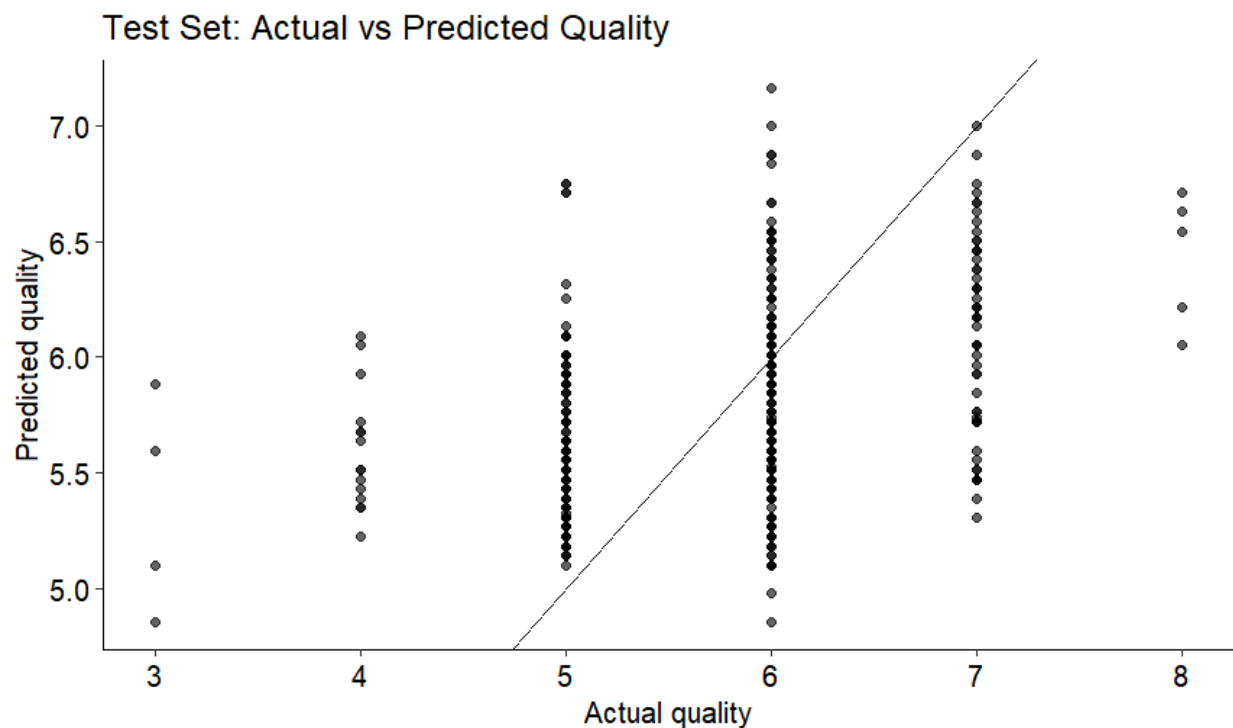


Figure 4. Test set plot of actual quality versus predicted quality from the final model, including the 45-degree reference line.

Prediction example (with intervals):

To demonstrate prediction, the final model was used to predict quality for a wine with alcohol content of 12.0% (within the dataset range of 8.4% to 14.9%). The predicted mean quality was:

$$\text{quality_hat} = 6.339$$

The 95% confidence interval for the mean predicted quality at 12.0% alcohol was [6.268, 6.410]. The 95% prediction interval for a single new wine at 12.0% alcohol was [5.141, 7.537]. The prediction interval is wider than the confidence interval because it accounts for the natural variability of individual wines in addition to uncertainty in the fitted regression line.

Summary of evaluation:

Overall, the final simple linear regression model shows that alcohol content is a statistically significant predictor of wine quality, and the model performs reasonably as a baseline. However, test-set accuracy and test R-squared indicate that alcohol alone does not provide strong predictive accuracy for individual wines. A more accurate approach would likely require additional predictors (multiple regression), but for the required simple regression model, this final model provides an interpretable relationship and a defensible prediction method.

Conclusion

In this project we used the UCI red wine dataset ($n = 1,599$) to model wine quality (0–10) as a function of alcohol content (%) using simple linear regression. After fitting on a 70/30 train–test split and removing 49 influential training observations using Cook’s distance, the final model was $\text{quality_hat} = 1.3817 + 0.4131(\text{alcohol})$. This implies that, on average, a 1% increase in alcohol content is associated with about a 0.413 increase in predicted quality score. The final model fit on the cleaned training data had $\text{RSE} = 0.609$ and $\text{R-squared} = 0.313$ (adjusted 0.312), so alcohol explains a meaningful but limited portion of quality variation. On the test set, performance was $\text{RMSE} = 0.706$, $\text{MAE} = 0.551$, and $\text{test R-squared} = 0.215$, which shows the relationship is real and positive but alcohol alone is not enough to predict individual wine quality with high accuracy. For example, at $\text{alcohol} = 12.0\%$, the model predicts a mean quality of 6.339 with a 95% confidence interval [6.268, 6.410] and a wider 95% prediction interval [5.141, 7.537]. A key limitation is that quality depends on many factors and is an integer rating, so a single linear predictor cannot capture all variability. Future work would compare other predictors and/or use multiple regression (and potentially nonlinear terms or interactions) with cross-validation to improve predictive accuracy.

Reference

<https://archive.ics.uci.edu/dataset/186/wine+quality>

sthda.com/english/articles/40-regression-analysis/167-simple-linear-regression-in-r/

https://medium.com/@ian.moore_83986/back-to-basics-linear-regression-with-r-931d22409d24

Appendix A. Key R Code:

The dataset was imported from a semicolon-delimited CSV file using `readr::read_delim()`. Data exploration was performed using `dim()`, `summary()`, `names()`, `head()`, `tail()`, and `view()`. Histograms were generated using `hist()` with customized labels and breaks, and multiple plots were arranged using `par(mfrow = ...)`. Correlation between predictors was assessed using `cor()`.

The linear regression model is reproducible because with `set.seed(456)`, `sample()` generates the same indices every time. The model was fitted on the training dataset using `lm()`; its summary was obtained with `summary()`. Confidence intervals for coefficients were computed using `confint()`. Additional formatted output tables were generated using `tidy()` and `glance()` from the broom package. Predictions were generated using `predict()`.

Diagnostic checks were performed using `plot(model)` to evaluate linearity and normality of residuals. A separate Q–Q plot was generated using `qqnorm()` and `qqline()` to assess normality visually.